# KPIT SPARKLE 2027

## THEME: WEARABLE ROBOTICS

## TITLE: Fail safe Steer-by-Wire Dynamic Force-Feedback Haptic Actuator with Active Collision-Rejection Virtual Wall & Road Impedance Synthesis

**TEAM NAME: TECHTONICS**

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# PROBLEM STATEMENT

### No Natural Force Feedback
In **Steer-by-Wire** systems, the mechanical connection is absent, so drivers do not feel natural steering feedback.

### Reduced Driver Awareness
Lack of physical feedback can make it difficult to judge road conditions, vehicle behaviour and nearby obstacles.

### Challenging in Critical Situations
Drivers may not get timely haptic cues during autonomous driving assistance or emergency manoeuvres.

### Safety and Confidence Concerns
Inadequate steering feedback can reduce driver confidence and may affect safe and intuitive vehicle control.

### Problem Statement
Develop a **Smart Steer-by-Wire Haptic Feedback System** that provides dynamic steering resistance and active obstacle avoidance assistance to give drivers a realistic, safe and intuitive driving experience in future electric and autonomous vehicles.

- Develop a Smart Steer-by-Wire Haptic Feedback System for EVs and autonomous vehicles.
- Integrate real-time steering angle, torque and vehicle speed sensors.
- Implement motor-driven force feedback for self-centering and speed-dependent steering resistance.
- Generate progressive counter-torque near obstacles to provide haptic steering assistance.
- Incorporate driver override, hardware protection and CAN-based real-time telemetry.

## NEED FOR THIS

**Lack of Natural Steering Feedback:** Steer-by-Wire systems eliminate direct mechanical connections, reducing the natural steering feel experienced by drivers.

**Improved Driver Awareness:** Drivers need physical feedback to better understand steering conditions and vehicle behaviour.

**Obstacle Avoidance Assistance:** Provides counter-torque feedback to alert drivers when steering towards potential hazards.

**Speed-Adaptive Steering:** Steering resistance must dynamically adjust according to vehicle speed for better control.

**Enhanced Driver Safety:** Haptic alerts help communicate lane departures, road conditions and critical driving situations.

**Driver Override Capability:** Ensures drivers can overcome electronic steering assistance during emergency manoeuvres.

# PROPOSED SOLUTION

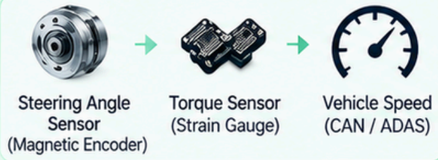
## Intelligent Haptic Steering Intervention

### 1 Smart Steer-by-Wire Haptic Feedback System

Develop an intelligent steering interface for electric and autonomous vehicles that provides dynamic haptic feedback to the driver.

### 2 Real-time Sensing

Integrate sensors to measure steering angle, driver-applied torque and vehicle speed for responsive control.



### 3 Motor-Driven Force Feedback

Use a motor-driven actuator to generate steering resistance, simulating natural steering feel with self-centering and speed-dependent stiffness.

### 4 Obstacle-Aware Steering Assistance

Generate progressive counter-torque when approaching obstacles to provide haptic alerts and assist the driver during critical situations.

### 5 Safety, Driver Override & CAN Telemetry

Incorporate driver override for emergency manoeuvres, hardware protection against faults and CAN-based telemetry for real-time monitoring and vehicle integration.

**Driver Override** (Emergency Control)

**Hardware Protection** (Current Limiting, Thermal Safety)

**CAN-based Telemetry** (Real-time Monitoring & Vehicle Communication)

## EXPECTED OUTCOMES

- Improved steering feel and driver awareness.
- Real-time obstacle-aware haptic assistance.
- Responsive steering with adaptive resistance.
- Enhanced driver control through override mechanisms.
- Reliable monitoring through CAN communication.

## KEY INNOVATIONS

**Virtual Impedance Control**
Dynamically adjusts steering stiffness and resistance based on vehicle speed and steering input.

**Progressive Haptic Obstacle Barrier**
Generates increasing counter-torque as the driver steers towards a detected obstacle.

**Intelligent Driver Override**
Detects intentional driver-applied torque and releases haptic resistance during emergency manoeuvres.

**Multi-Sensor Feedback Integration**
Combines steering angle, torque and vehicle-speed data for responsive steering assistance.

**Road Texture Simulation**
Generates distinct vibration patterns to simulate road irregularities and lane-departure warnings.

**Real-Time CAN Telemetry**
Enables continuous monitoring of steering parameters, actuator conditions and system status.

# SYSTEM ARCHITECTURE

## 1 Hardware Layer (Sensing, Processing & Actuation)

### Sensors & Inputs

- **Steering Angle Sensor** (Magnetic Encoder) AS5600 → Steering Angle (θ)
- **Torque Sensor** (Strain Gauge + Amplifier) → Driver Torque (T_driver)
- **Vehicle Speed** (CAN / ADAS) → Vehicle Speed (v)
- **Obstacle Sensor** (Ultrasonic / Radar / (Camera) → Obstacle Info (distance, lane)

### Embedded Controller

**MCU** (STM32 / NXP)

- Sensor data acquisition
- Signal processing
- Haptic control algorithm
- Driver override logic
- CAN communication
- System diagnostics

Torque Command

### Motor Driver & Actuation

**MOSFET H-Bridge Motor Driver**

- Bidirectional torque control
- Current sensing
- Overcurrent protection
- Thermal protection

Controlled Current

**Steering Motor** (High-Torque, Coupled to Steering Wheel)

Feedback (Current, Temperature)

## 2 Software Layer (Control & Intelligence)

**Sensor Data Processing**
- Filtering & Calibration
- Sensor Fusion

**Haptic Control Algorithm**
- Self-centering control
- Speed-dependent stiffness
- Obstacle counter-torque
- Road texture simulation

**Driver Override Detection**
- Intent recognition
- Threshold-based override

**Safety & Fault Handling**
- Overcurrent / overtemperature
- Sensor fault detection
- Fail-safe mode

**Real-Time Tasks** (FreeRTOS)

**CAN Communication**
- Tx/Rx steering data
- Vehicle integration
- Diagnostics

**Data Logging**
Store events and parameters

**Monitoring & GUI** (PC / Dashboard)
- Real-time visualization
- Parameter tuning

## 3 Output & User Interaction

**Haptic Steering Feedback**

- Variable steering resistance
- Self-centering feel
- Progressive counter-torque
- Vibration alerts (road / lane)

**Driver**

- Feels natural steering feedback
- Receives haptic alerts
- Can override in emergencies

## 4 Power Supply

12V Vehicle Supply (or EV Supply)

DC-DC Converter (5V / 3.3V)

- MCU & Sensors
- Motor Driver
- Communication Modules

## 5 Communication Layer (CAN Bus)

Embedded Controller (MCU) ⟷ CAN Tx/Rx ⟷ **CAN Bus** ⟷ CAN Tx/Rx ⟷ Vehicle ECU / ADAS

**Transmitted Data**
- Steering angle, torque, speed
- Obstacle information
- System status and diagnostics

## 6 Monitoring & Diagnostics (PC / Cloud)

Real-Time Dashboard (PC Application) ⟷ Cloud / Data Logging (Optional)

- Real-time visualization
- Parameter tuning
- Event logging
- Diagnostics and fault codes
- Performance analysis

**SYSTEM WORKFLOW:**

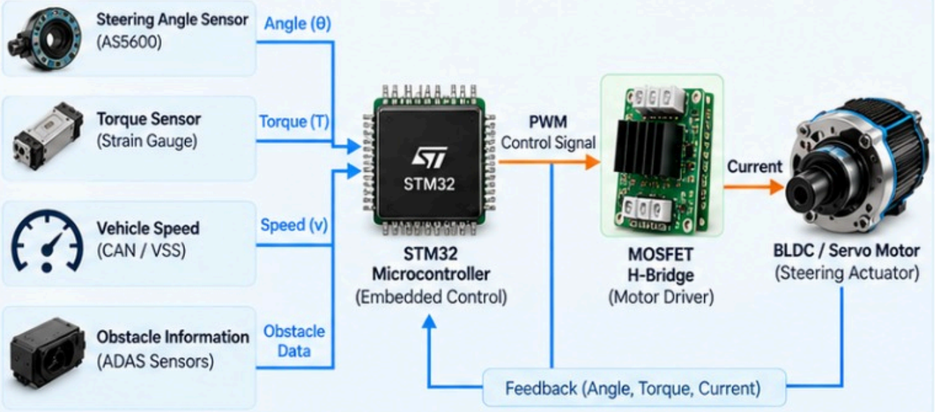
Sensor Inputs → Embedded Controller → Haptic Control Algorithm → Motor Driver → Steering Feedback

# HARDWARE DESIGN

## EMBEDDED CONTROL AND ACTUATION SUBSYSTEM

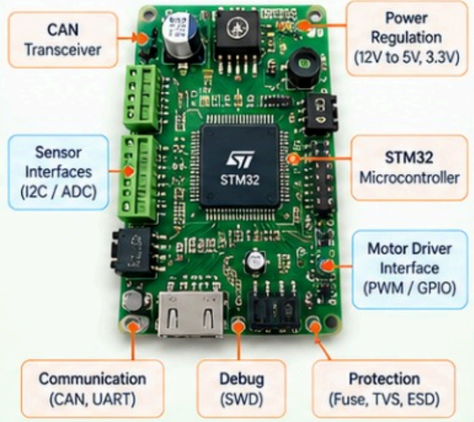
### 1 Hardware Overview

Embedded control unit with sensors, motor driver and steering actuator



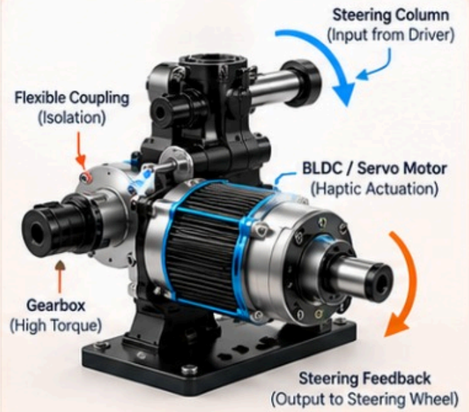

### 2 Embedded Control Unit

STM32 based control und signal processing




### 3 Steering Actuation Subsystem

Motor driven haptic feedback mechanism




### 4 Sensors and Components

Key hardware elements used in the prototype

**Steering Angle Sensor AS5600**
- Magnetic encoder
- 0–360° measurement
- I2C interface

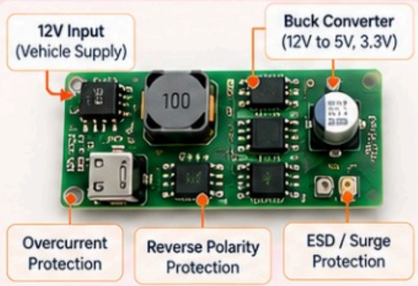
**Torque Sensor (Strain Gauge)**
- Measures driver torque
- High sensitivity
- Analog output

**BLDC / Servo Motor**
- Haptic feedback
- Variable resistance
- High torque output

**MOSFET H-Bridge Motor Driver**
- PWM motor control
- High current rating
- Current feedback

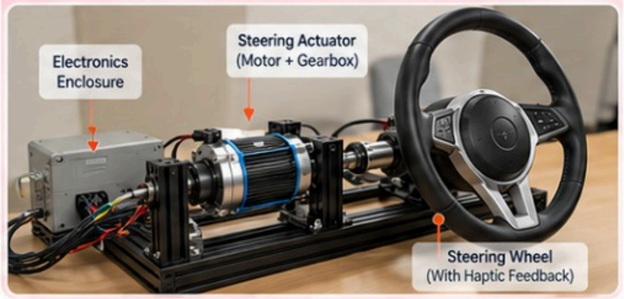
### 5 Power and Protection Circuitry

Stable and reliable operation




### 6 Prototype Hardware Setup

Integrated steering mechanism with electronics




**Component Summary**

- STM32 Microcontroller
- AS5600 Magnetic Encoder
- Strain Gauge Torque Sensor
- BLDC / Servo Motor Haptic Actuation
- MOSFET H-Bridge Motor Driver
- CAN Transceiver Vehicle Communication
- Protection Circuitry Overcurrent / ESD

## KEY HARDWARE COMPONENTS

STM32 | AS5600 | Strain Gauge | INA128 | MOSFET H-Bridge | DC Geared Motor | INA240 | CAN Transceiver

**Real-Time Sensor Acquisition:**
Captures steering angle, driver-applied torque and vehicle speed for continuous monitoring.

**Embedded Control Processing:**
Processes sensor inputs and calculates the required steering feedback torque.

**Bidirectional Motor Control**:
Uses a MOSFET-based H-Bridge to control motor direction, speed and torque.

**Closed-Loop Actuation:**
Continuously adjusts motor output based on steering position and torque feedback.

**Obstacle-Based Intervention:**
Generates controlled counter-torque when the steering input approaches a simulated obstacle boundary.

# CONTROL ALGORITHM AND WORKING PRINCIPLE

## PROGRESSIVE HAPTIC INTERVENTION AND DRIVER OVERRIDE

### 1 Control Algorithm Flow

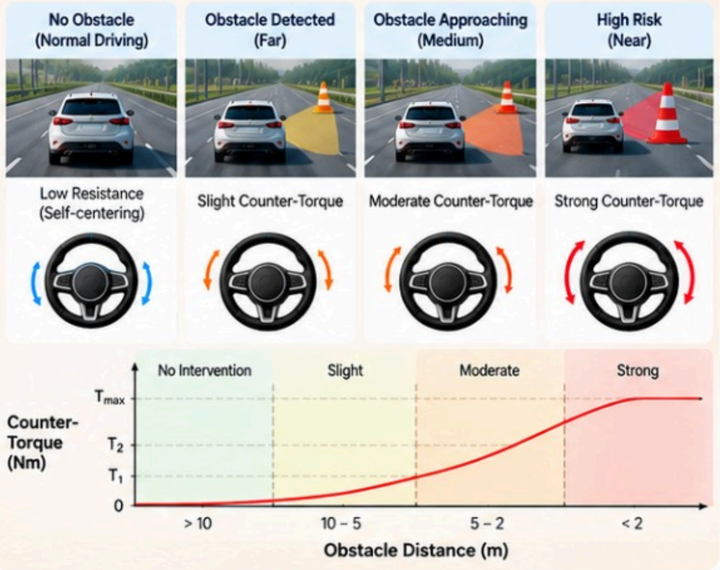
From sensor inputs to haptic steering output

1. **Sensor Acquisition**
   Steering angle, torque, vehicle speed, obstacle data
2. **Signal Processing**
   Filtering, sensor fusion, state estimation
3. **Driver Intent Estimation**
   Detect intentional steering input
4. **Haptic Control Logic**
   Virtual impedance control
   Progressive counter-torque
5. **Safety & Override Check**
   Driver override, fault monitoring
6. **Motor Driver Output**
   PWM generation and current control

Feedback (Angle, Torque, Current)

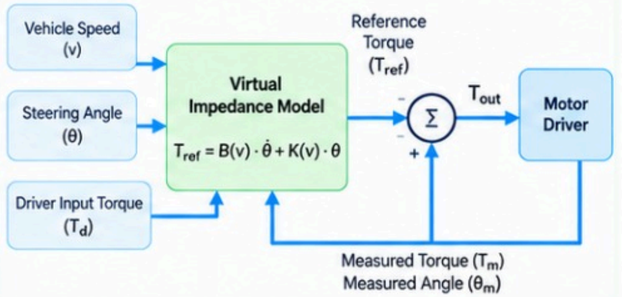
### 2 Progressive Haptic Intervention

Counter-torque increases with obstacle proximity




### 3 Virtual Impedance Control

Speed-adaptive steering resistance




**Speed-based Impedance Parameters**

| Vehicle Speed (km/h) | Damping B (Nm·s/rad) | Stiffness K (Nm/rad) |
|---|---|---|
| 0 – 20 | Low | Low |
| 20 – 60 | Medium | Medium |
| > 60 | High | High |

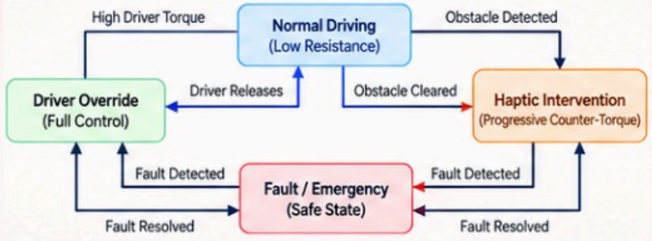
### 4 Driver Override Mechanism

Driver input is always given priority

- Detects intentional high torque input
- Reduces haptic intervention immediately
- Allows complete driver control
- Resumes intervention when condition is safe

### 5 Control States and Switching Logic

Adaptive intervention based on driving condition




### 6 Safety and Fault Handling

Fail-safe operation

- **Overcurrent / Overload Protection**
  Limits motor current and disables output
- **Over-temperature Protection**
  Monitors driver and motor temperature
- **Sensor Fault Detection**
  Checks sensor validity and signal range
- **CAN Communication Monitoring**
  Detects loss of vehicle data
- **Safe State Activation**
  Disables haptic intervention in fault condition

## ADAPTIVE CONTROL PARAMETERS

- **Steering stiffness (K):** Controls steering resistance.
- **Damping coefficient (B):** Controls resistance to steering movement.
- **Counter-torque (T):** Determines obstacle intervention strength.
- **Override threshold:** Identifies intentional driver intervention.
- **PWM duty cycle:** Controls motor output.

## PERFORMANCE EVALUATION PARAMETERS

- **Steering response time.**
- **Torque response and accuracy.**
- **Obstacle detection response.**
- **Driver override response time.**
- **Motor current and temperature.**
- **Control-loop execution time.**

## CLOSED-LOOP CONTROL

Sensor Feedback → Control Algorithm → Motor Actuation → Torque Measurement → Continuous Correction

# DEMONSTRATION AND OPERATING SCENARIOS

## REAL-TIME ADAS-TO-STEERING INTERACTION

### DEMO DASHBOARD SCREENSHOT

KPIT

**STEER-BY-WIRE HAPTIC ACTUATOR**

Automotive Chassis Domain Controller - ISO 26262 & UN R79 SIL Telemetry

NORMAL MODE

T: 00:04.13

**PHASE 1/5: BASE ROAD IMPEDANCE & SPRING CENTERING**

Testing dynamic self-centering (K_p * theta) and compliant feel

**STEERING HMI DIGITAL TWIN**

-31.0 deg

SPRING CENTERING TORQUE
+1.09 Nm

BARRIER COUNTER-TORQUE
+0.00 Nm

NET MOTOR TORQUE COMMAND (tau_cmd)
+1.09 Nm

**ADAS SCENARIO & ROAD SIMULATION**

VEHICLE SPEED (v_veh): 40 km/h

HUMAN EFFORT TORQUE: 2.5 Nm

SAFETY: ISO 26262 ASIL-D ACTIVE

Compliant steering feel; torque clamped to 5.5 Nm.

**TELEMETRY HARVEST & CAN-FD MONITOR**

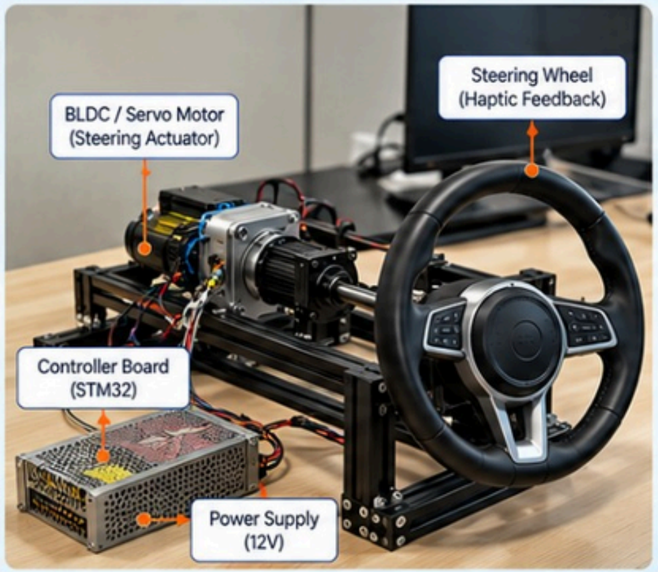
-- Angle (deg) -- Motor Torque (Nm) -- Human Effort (Nm)

| CAN-ID | DLC | PAYLOAD (HEX) | CYCLE |
|---|---|---|---|
| 0x102 | 8 | 36 15 00 00 FF AA | 20 ms |
| 0x104 | 8 | 36 15 00 00 FF AA | 20 ms |
| 0x220 | 4 | 36 15 00 00 FF AA | 50 ms |
| 0x350 | 8 | 36 15 00 00 FF AA | 100 ms |

CURRENT: 1.38 A | TEMP: 35.4 C

PWM DUTY: 19.7 % | LOOP JITTER: 0.18 ms

### PROTOTYPE SETUP




### TESTING SCENARIOS

Low resistance
(Self-centering)

Slight counter-torque

Moderate counter-torque

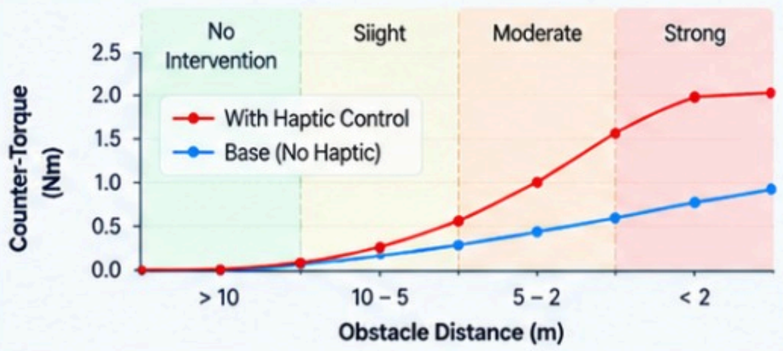
Strong counter-torque,
haptic warning

# RESULTS AND ANALYSIS

## PROTOTYPE DEVELOPMENT, COMPONENTS AND VALIDATION

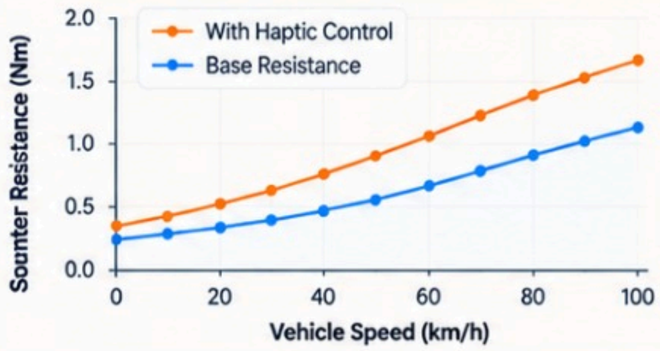
### 1 Steering Torque Response

Counter-torque vs. obstacle distance



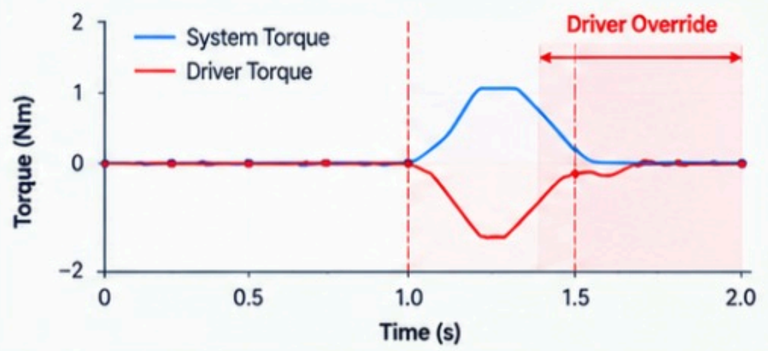

### 2 Speed-based Steering Resistance

Steering effort vs. vehicle speed




### 3 Driver Override Response

System vs. driver applied torque




### 4 Experimental Results Summary

Key performance metrics

| Parameter | Base System | With Haptic Control | Improvement |
|---|---|---|---|
| Max Counter-Torque (Nm) | 0.8 | 2.0 | +150% |
| Steering Resistance @ 100 km/h (Nm) | 1.1 | 1.7 | +55% |
| Driver Override Time (ms) | – | < 100 | – |
| Control Latency (ms) | – | 45 | – |

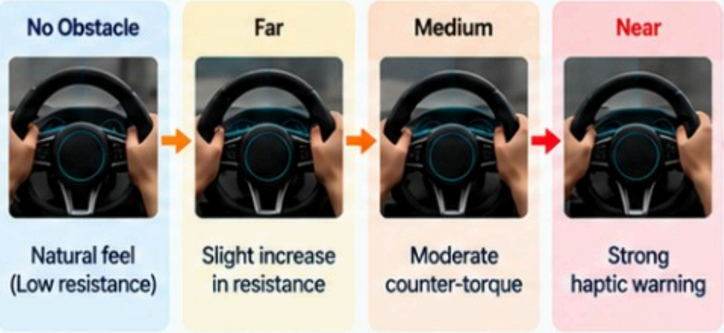
### 5 Scenario-wise Performance

System behaviour under different conditions

| Scenario | Obstacle Distance | Steering Response | Result |
|---|---|---|---|
| Normal Driving | > 10 m | No intervention (Natural feel) | Smooth Operation |
| Obstacle Ahead (Far) | 5 – 10 m | Slight counter-torque | Early Warning |
| Obstacle Approaching (Medium) | 2 – 5 m | Moderate counter-torque | Effective Alert |
| Critical Proximity (Near) | < 2 m | Strong counter-torque | Clear Warning |

### 6 User Feel and Steering Feedback

Progressive and natural haptic feel




### 7 Discussion

Interpretation of results

**Progressive Torque Response**
Counter-torque increases with obstacle proximity

**Speed Adaptation**
Steering resistance scales with vehicle speed

**Reliable Override**
Driver input is always prioritized

**Stable Operation**
Consistent performance across all test scenarios

### 8 Key Findings

Major outcomes from prototype testing

- Effective Obstacle-aware haptic intervention
- Smooth and progressive steering response
- Fast and reliable driver override
- Stable and consistent system operation

# IMPACT AND FUTURE SCOPE

## TOWARDS SAFER HUMAN–MACHINE INTERACTION IN SOFTWARE-DEFINED VEHICLES

### 1 Real-World Applications

Enhancing safety in everyday driving scenarios

**Urban Driving**
Obstacle warning in congested traffic

**Highway Driving**
Lane-side obstacle awareness

**Parking Assistance**
Close-range obstacle warning

**Low Visibility Conditions**
Additional haptic safety layer

### 2 Key Impacts

Safer, smarter and more intuitive driving

**Improved Road Safety**
Reduces collision risk through haptic alerts

**Intuitive Driver Awareness**
Guides driver without visual distraction

**Supports ADAS**
Complements camera and radar systems

**Enhanced Driving Experience**
More confident and comfortable control

**Broader Applicability**
Suitable for both personal and commercial vehicles

**Potential for Mass Adoption**
Low-cost and easy integration with existing steering systems

### 3 Feasibility and Cost Analysis

Practical, low-cost and scalable solution

| Component | Estimated Cost (INR) | Availability | Remarks |
|---|---|---|---|
| BLDC / Servo Motor | 4,000 – 6,000 | Readily available | Automotive grade |
| Controller (STM32) | 800 – 1,200 | Readily available | Low-cost MCU |
| Sensors (Angle, Current) | 1,000 – 1,500 | Readily available | Automotive sensors |
| Motor Driver | 800 – 1,200 | Readily available | High current driver |
| Power Supply (12V) | 500 – 800 | Readily available | Automotive DC supply |
| Miscellaneous (PCB, etc.) | 1,000 – 1,500 | Readily available | Prototype build |
| **Total (Prototype)** | **8,100 – 12,300** | – | – |

**Technically Feasible**
Uses commercially available components

**Cost-Effective**
Suitable for prototype and real-world scaling

**Easy Integration**
Can be adapted to existing steering systemts

**Scalable**
Applicable to multiple vehicle segments

## FUTURE SCOPE

- **Integration with Advanced ADAS:** Combine with lane keeping and automatic braking systems.
- **AI-Based Predictive Haptic Control:** Learn driver behaviour and road patterns.
- **Vehicle-to-Everything (V2X) Support:** Provide haptic alerts using connected infrastructure.
- **Miniaturization and Automotive-Grade Design:** Optimize for production and real-world deployment.
- **Testing in Real Vehicles:** Validate under real driving conditions.
- **Extension to Autonomous Vehicles:** Develop a haptic interface for human-in-the-loop control.

A practical, low-cost and impactful solution to improve driving safety through intelligent haptic feedback!

**GITHUB LINK: CLICK HERE**

**DEMO VIDEO: CLICK HERE**

# THANK YOU!

"The future of driving isn't just about controlling the wheel; it's about making the wheel communicate with the driver."